Chapter 14, Part 2

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Money Creation and Central Banks

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Roadmap

1. Money creation and the deposit multiplier
2. The monetary base and money supply
3. The Federal Reserve: structure and tools
4. The ECB and the evolution of U.S. banking

Determining the Money Supply

Money supply = public currency + checkable deposits

Simplified measure: M = C + D

The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing	$0	$1,000	$1,000

The deposit changes the form of money, not its total.

How Banks Create Money

Starting wages: $1,000 cash · Model reserve requirement: rr = 10%



1. Deposit



2. Loan → cash



3. Spend



4. Redeposit

The supplier deposits $900 at Bank B.

Lending adds money; payment and redeposit do not add it again.

Money-supply accounting ($)

Step	C	D	M
Before	1,000	0	1,000
Deposit	0	1,000	1,000
Loan credit	0	1,900	1,900
Cash / spend	900	1,000	1,900
Redeposit	0	1,900	1,900

D: deposits · C: public currency

M = C + D

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

At Bank A: $1,000 − (0.10 × $1,000) = **$900**.

rr = 10% · Each row follows lending and payment to the next bank.

Bank	Deposits	Reserves kept	New loan
A	$1,000	$100	$900
B	$900	$90	$810
C	$810	$81	$729
…	…	…	…

The Simple Money Multiplier

Initial cash deposit: \$1,000 · rr = 0.10

$$D = 1{,}000 + 1{,}000(1 - rr) + 1{,}000(1 - rr)^2 + \ldots$$

$$D = \frac{1{,}000}{rr} = \frac{1{,}000}{0.10} = \mathbf{\$10{,}000}$$

\$10,000 total deposits includes the original \$1,000.

\$9,000 additional deposits are created through lending.

Under these assumptions, \$1 of reserves supports \$10 of deposits.

Net increase in C + D from before the cash deposit: \$9,000.

Why Reality Is More Complicated

Full redeposit. People redeposit all loan proceeds.

No excess reserves. Banks lend the amount allowed by the model.

Borrowers qualify. Willing, creditworthy borrowers want the loans.

More public cash or excess reserves → less deposit expansion.

1/rr is a teaching model, not an automatic prediction.
The 10% rule is hypothetical; U.S. reserve requirements have been 0% since March 2020.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Monetary Base and Money Supply

Monetary base

$$MB = C + R$$

Public currency + bank reserves

Reserves = vault cash + balances at the Fed.

Money supply

$$M = C + D$$

Public currency + checkable deposits

Bank reserves are not public deposits.

$$\text{Money multiplier} = \frac{M}{MB}$$

Next: how Fed operations change reserves and the monetary base.

Money Creation Checkpoint

$1,000 of existing public cash · rr = 10% · Full lending and redeposit

1. Cash becomes a deposit.

Does the money supply rise immediately?

No. Currency falls $1,000; deposits rise $1,000.

2. Lending repeats across banks.

What are total deposits and the net increase in money?

Deposits: $10,000. Net increase in M: $9,000.

3. Banks make loans.

Does lending itself create monetary base?

No. It creates deposits; it does not create base money.

The Federal Reserve System



1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

Reserve Bank locations. San Francisco also serves Alaska and Hawaii.

Board of Governors

7 seats · Washington, D.C.

Nominated by the President; confirmed by the Senate.

Oversees the System. Governors serve staggered 14-year terms.

Federal Reserve, System Structure · Map: Natural Earth

The Federal Open Market Committee

FOMC = the Fed's monetary-policy committee.

7	+	1	+	4
Board governor seats		New York Fed president		Rotating Reserve Bank presidents

Sets the **target range for the federal funds rate** and directs open-market policy.

All Reserve Bank presidents participate; voting rights rotate.
The FOMC does not set every household or business loan rate.

Federal Reserve, FOMC · Federal Reserve, System Structure

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

FFR: the federal funds rate—the interest rate in this market.

Policy tool	What changes
Discount rate	Cost of borrowing from the Fed
Interest on reserve balances	Return on eligible balances at the Fed
Open-market operations	Securities and bank reserve balances

Next week: interest rates, spending, and aggregate demand.

New York Fed, Monetary Policy Implementation

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Lower discount rate:
Cheaper Fed funding; borrowing becomes more attractive.

Higher discount rate:
Costlier Fed funding; borrowing becomes less attractive.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

A reservation rate:
a safe alternative return to lending.

Higher IORB raises that benchmark; lower IORB reduces it.

IORB influences lending rates; it is not a fixed minimum rate on every loan.

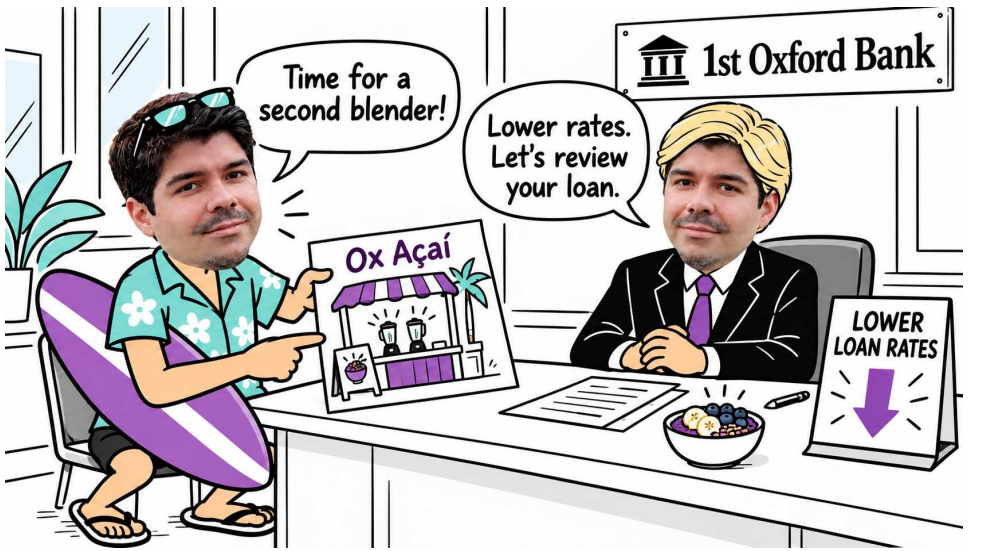
Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

Open-Market Purchases

The Fed buys a $1,000 Treasury security from a commercial bank.



Purchase: reserves enter the bank.



Easier financing can support borrowing.

Bank reserves +$1,000 · Monetary base +$1,000

Bank securities −$1,000 · Public deposits: no immediate change.

Loan-rate effects are conditional; reserve changes alone do not guarantee them.

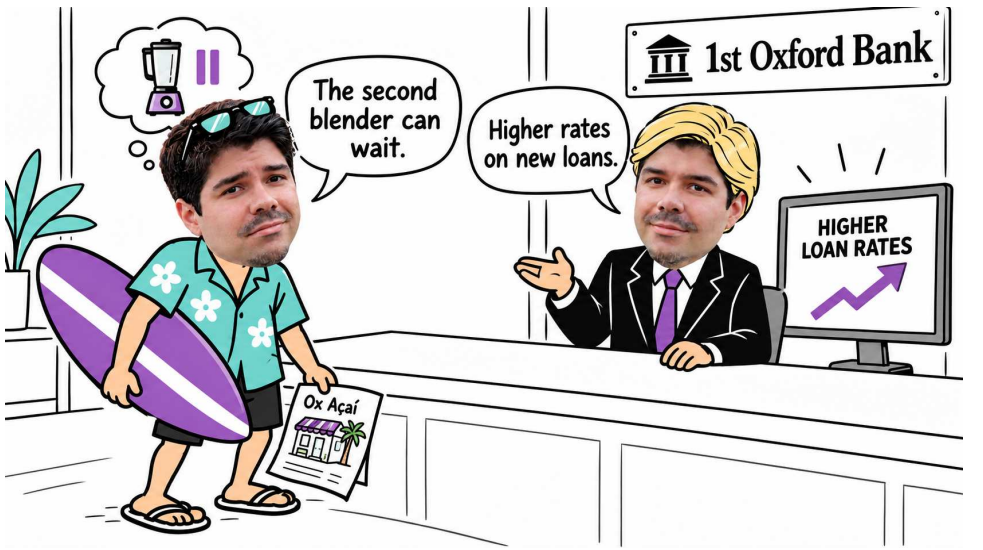
New York Fed, Monetary Policy Implementation

Open-Market Sales

The Fed sells a $1,000 Treasury security to a commercial bank.



Sale: reserves leave the bank.



Costlier financing can postpone borrowing.

Bank reserves −$1,000 · Monetary base −$1,000

Bank securities +$1,000 · Public deposits: no immediate change.

Loan-rate effects are conditional; reserve changes alone do not guarantee them.

New York Fed, Monetary Policy Implementation

The ECB and the Federal Reserve

ECB = European Central Bank · Eurosystem = ECB + euro-area national central banks.

	Federal Reserve	ECB / Eurosystem
Central board	7 Board governor seats	6 Executive Board members
Policy committee	FOMC	Governing Council
Operating network	12 regional Reserve Banks	Euro-area national central banks
Main policy goals	Maximum employment; stable prices	Price stability is primary

Both use policy rates, lending facilities, and asset operations.

The ECB’s deposit facility offers a comparable remuneration benchmark.

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

Banking Panics and the Creation of the Fed

1907: banking panic

Withdrawal demands outran available cash; emergency support depended on private coordination.

1913: the Federal Reserve Act

A central reserve system could supply liquidity when individual banks could not.

1930–1933: renewed bank runs

The Fed did not prevent a severe monetary contraction. The 1933 Banking Act created federal deposit insurance.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

The Savings-and-Loan Crisis

1980s: a funding mismatch

Thrifts held long-term, fixed-rate mortgages while rising rates increased their funding costs.

Risk-taking and weak oversight

Broader investment powers, weak capital and insured deposits encouraged costly risk-taking.

1989: cleanup and reform

FIRREA reorganized thrift oversight and resolution. Public funds helped cover the failed institutions' losses.

Fed History: Savings-and-Loan Crisis · FDIC: 1980–1989

The 2008 Crisis and Its Aftermath

2007–2009: housing losses and funding runs

Mortgage-related losses and withdrawals of short-term funding spread stress beyond deposit banks.

2008: emergency support

The Fed expanded liquidity lending and asset purchases; the Treasury provided capital support.

2010: Dodd–Frank reforms

New systemic-risk oversight and consumer protection addressed gaps exposed by the crisis.

Fed History: Great Recession · Federal Reserve: Dodd–Frank